King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 201 Major Exam II Term 241

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. If $\tan(x+y)+\cos z = 2$, then $\dfrac{\partial z}{\partial x}+\dfrac{\partial z}{\partial y} =$

(a) $\dfrac{2\tan(x+y)}{\cos z}$

(b) $\dfrac{\tan^2(x+y)}{\cos z}$

(c) $\dfrac{\csc^2(x+y)}{\sin z}$

(d) $\dfrac{2\sec^2(x+y)}{\sin z}$

(e) $-\dfrac{\sec^2(x+y)}{\sin z}$

2. Consider the following statements about the function

$$f(x,y,z) = \frac{xy+yz^2+xz^2}{x^2+y^2+z^2}$$

(I) $\lim_{(x,y)\to(0,0)} f(x,y,0) = 0$, (II) $\lim_{(x,z)\to(0,0)} f(x,0,z) = 0$, (III) $\lim_{x\to 0} f(x,x,x) = \dfrac{1}{3}$. Then

(a) only (I) is true

(b) (II) and (III) are true

(c) only (III) is true

(d) (I) and (III) are true

(e) (I) and (II) are true

3. An equation of the plane passing through the point $(1,2,3)$ and parallel to the xy-plane is

(a) $x = 1$

(b) $x+y-3 = 0$

(c) $z = 3$

(d) $x+y+z-6 = 0$

(e) $y = 2$

4. The level curve of the function $f(x,y) = \sqrt{9-x^2-y^2}$ passing through the point $(2, \sqrt{5}, f(2,\sqrt{5}))$ is a circle of radius:

(a) 2

(b) $\sqrt{5}$

(c) 9

(d) 3

(e) 1

5. The range of $f(x,y) = \sqrt{4-x^2-y^2}$ is

(a) $[0,2]$

(b) $(0,4)$

(c) $(-\infty,\infty)$

(d) $[2,\infty)$

(e) $[0,\infty)$

6. $\lim_{(x,y)\to(0,0)}(x^2+y^2)\ln(x^2+y^2) =$

(a) $\ln 2$

(b) 0

(c) 2

(d) ∞

(e) 1

7. If $f(x,y,z) = e^{-x}\sin(yz)$, then $f_{xyy} =$

(a) $-e^{-x}z^2\cos(yz)$

(b) $e^{-x}\cos(yz)$

(c) $e^{-x}y^2\cos(yz)$

(d) $-e^{-x}\sin(yz)$

(e) $e^{-x}z^2\sin(yz)$

8. An equation of the tangent plane to the surface $z^2-2x^2-2y^2 = 12$ at the point $(1,-1,4)$ is given by $x+ay+bz+c=0$, then $a+b+c =$

(a) 9

(b) 3

(c) 1

(d) 12

(e) 6

9. The surface given by the equation $x - y^2 - 4z^2 = 2$ is:

(a) Ellipsoid

(b) Hyperboloid of two sheets

(c) Elliptic cone

(d) Hyperboloid of one sheet

(e) Elliptic paraboloid

10. The directional derivative of $f(x, y) = e^y \sin x$ at the point $P(0, 0)$ in the direction of the vector $\overrightarrow{PQ}$, where $Q(2, 1)$, is equal to

(a) 2

(b) $\dfrac{2}{\sqrt{5}}$

(c) $\dfrac{3}{\sqrt{5}}$

(d) $-\sqrt{5}$

(e) $\dfrac{1}{\sqrt{5}}$

11. Let $w = x^2 - y^2$, where $x = s\cos t$ and $y = s\sin t$. The value of $\dfrac{\partial w}{\partial t}$ when $s = 3$ and $t = \dfrac{\pi}{4}$ is equal to

(a) -12

(b) 6

(c) 3

(d) 9

(e) -18

12. The distance between the planes given by the equations

$$-x + 6y + 2z = 3 \quad \text{and} \quad -\frac{1}{2}x + 3y + z = 4$$

is

(a) $\dfrac{7}{\sqrt{41}}$

(b) $\dfrac{3}{\sqrt{41}}$

(c) $\dfrac{8}{\sqrt{41}}$

(d) $\dfrac{1}{\sqrt{41}}$

(e) $\dfrac{5}{\sqrt{41}}$

13. Using total differential, the expression $5\sqrt{(4.03)^2 + (3.1)^2} - 5\sqrt{4^2 + 3^2}$ is approximately equal to

(a) 0.60

(b) 0.50

(c) 0.35

(d) 0.30

(e) 0.42

14. A set of symmetric equations for the tangent line to the curve of intersection of the surfaces

$$x^2 + y^2 + z^2 = 14 \quad \text{and} \quad x - y - z = 0$$

at the point $(3, 1, 2)$ is

(a) $\dfrac{x-3}{4} = y - 1 = \dfrac{z-3}{5}$

(b) $x - 3 = \dfrac{y-1}{5} = \dfrac{z-2}{-4}$

(c) $x - 3 = \dfrac{y-1}{-5} = \dfrac{z-2}{4}$

(d) $\dfrac{x-3}{2} = \dfrac{y-1}{10} = \dfrac{z-2}{-7}$

(e) $\dfrac{x-3}{2} = \dfrac{y-1}{5} = \dfrac{z-2}{-4}$

15. Consider a line that passes through the point $(5, -3, -4)$ and is parallel to the vector $\vec{u} = \langle 2, -1, 3 \rangle$. If this line also passes through the point $(7, -4, a)$, then a is equal to

(a) -3

(b) 4

(c) -2

(d) -1

(e) 0

Answer Key

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Q	Ans	Brief explanation
1	**(d)**	$z_x = z_y = -\frac{\sec^2(x+y)}{-\sin z}$, so the sum is $\frac{2\sec^2(x+y)}{\sin z}$.
2	**(b)**	(I) path-dependent (0 vs $\frac{1}{2}$), (II) $\le \lvert x\rvert \to 0$, (III) $\frac{1+2x}{3} \to \frac{1}{3}$.
3	**(c)**	Normal $\langle 0,0,1\rangle$ through $(1,2,3)$: $z = 3$.
4	**(d)**	$f(2,\sqrt{5}) = 0$, so the level curve is $x^2 + y^2 = 9$, radius 3.
5	**(a)**	$4 - x^2 - y^2 \in [0,4]$ on the domain, so $f \in [0,2]$.
6	**(b)**	$u = x^2 + y^2 \to 0^+$: $u \ln u = \frac{\ln u}{1/u} \to 0$ by L'Hopital.
7	**(e)**	$f_x = -e^{-x}\sin yz$, $f_{xy} = -ze^{-x}\cos yz$, $f_{xyy} = z^2 e^{-x}\sin yz$.
8	**(b)**	$\nabla F = \langle -4,4,8\rangle$: plane $x - y - 2z + 6 = 0$, so $a + b + c = 3$.
9	**(e)**	$x = 2 + y^2 + 4z^2$: one linear variable, two squares of the same sign, so elliptic paraboloid.
10	**(b)**	$\nabla f(0,0) = \langle 1,0\rangle$, $\mathbf{u} = \frac{\langle 2,1\rangle}{\sqrt{5}}$: $D_{\mathbf{u}} f = \frac{2}{\sqrt{5}}$.
11	**(e)**	$w_t = -2s^2 \sin 2t$, so at $s = 3$, $t = \frac{\pi}{4}$: -18.
12	**(e)**	Parallel planes $-x + 6y + 2z = 3$ and $= 8$: distance $\frac{5}{\sqrt{41}}$.
13	**(e)**	$f = 5\sqrt{x^2 + y^2}$ at $(4,3)$: $df = 4(0.03) + 3(0.1) = 0.42$.
14	**(b)**	$\langle 6,2,4\rangle \times \langle 1,-1,-1\rangle = \langle 2,10,-8\rangle \parallel \langle 1,5,-4\rangle$.
15	**(d)**	$x = 5 + 2t = 7 \Rightarrow t = 1$ (and $y = -4$ checks), so $a = -4 + 3 = -1$.